IPCON® PPA GF FR

Glass Fiber Reinforced Flame Retardant Polyphthalamide Filament

This is the filament's Technical Data Sheet (TDS). If you want the Material Safety Data Sheet (MSDS), please contact the customer service to download.

• Product Information

Item	Data
Filament Diameter	1.75 ± 0.05 mm
Net Filament Weight	0.75 kg ± 0.5%
Filament Length	235 m ± 2%
Spool Material	Cardboard (max heat resistance: 150 °C)
Spool Size	Outer Cardboard's diameter: 200 mm; Inner roller's diameter: 110 mm
Recommendation for body protection during use	This material contains glass fiber, which may cause skin irritation. So it is recommended to wear gloves throughout the operation.

• Filament's Properties

Physical Properties	Testing Methods	Data
Density	ISO 1183-3	1.32 g/cm³
Shrinkage Rate*	Specially-defined	0.17% - 0.43%
Surface Hardness	Shore hardness	88D
Saturated Water Absorption Rate	25 °C, 55% RH, room air	1.29%
Melt Index	290 °C, 2.16 kg	17.8 g/10 min
Melting Temperature	DSC, 10 °C/min	251 °C
Vicar Softening Temperature	ISO 306, GB/T 1633	223 °C
Heat Deflection Temperature	ISO 75, 0.45 MPa	214 °C
Relative Thermal Index (RTI) Max Continuous Service Temperature	Actual working conditions simulation test	110 - 130 °C (dry state) (Related to load, size, shape, wall thickness, rate of water content, etc., and the value after annealing is higher than that before annealing in general)

Mechanical Properties*	Testing Methods	Data*
Tensile Strength XY	ISO 527, GB/T 1040	73 MPa
Tensile Strength Z	ISO 527, GB/T 1040	38 MPa
Young's Modulus XY	ISO 527, GB/T 1040	7352 MPa
Young's Modulus Z	ISO 527, GB/T 1040	3896 MPa
Breaking Elongation Rate XY	ISO 527, GB/T 1040	8.2%
Breaking Elongation Rate Z	ISO 527, GB/T 1040	5.6%

Flexural Strength XY	ISO 178, GB/T 9341	134 MPa
Flexural Strength Z	ISO 178, GB/T 9341	62 MPa
Flexural Modulus XY	ISO 178, GB/T 9341	6346 MPa
Flexural Modulus Z	ISO 178, GB/T 9341	4168 MPa
Impact Strength XY	ISO 179, GB/T 1043	22.3 kJ/m²
Impact Strength Z	ISO 179, GB/T 1043	5.9 kJ/m²

*The shrinkage rate and mechanical properties were tested on specimens printed by IPCON using a Bambu P1S printer (nozzle temperature: 290 °C, bed temperature: 100 °C, printing speed: 100 mm/s) and IPCON PPA GF FR (Black) filament. The parameters of specimens of mechanical properties are as following: infill density—100%, infill pattern—concentric; and those of the 2 models of shrinkage (2 crosses that are vertical in the X and Y directions with designed lengths of 100.00 and 150.00 mm respectively) are as following: wall loop—2 and 6, infill density—15% and 25%, infill pattern—grid. The data is for user reference and comparison only. When using it, users are responsible for the safety, legal compliance and product performance.

• Main Features and Application Scenarios

1. Top-level flame retardation, excellent heat insulation and resistance (HDT 214 °C @ 0.45 MPa, breakdown voltage up to 19 kV), super rigidity, high strength, low warpage, widely compatible with mainstream consumer printers (printing temperature below 300 °C), 4 basic colors: black, white, orange and blue.
2. Can be used to print: A. parts for automobiles, electronic and electrical equipment, which are supposed to be flame-retardant, temperature-resistant, moisture-resistant, and insulating; B. tools and fixtures with high rigidity, high strength, and low warpage; C. functional prototypes with top-level mechanical and safety performance.

• Recommended Printing Settings

Drying Settings Before Printing	100 - 130 °C, 10 - 14 h (Blast drying oven)
Printing Temperature and Humidity	≤ 120 °C, ≤ 20% RH (Sealed with desiccant)
Storage Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Compatible Support Material	Itself or support filament for PA
Compatible Printer Type	Enclosed-frame (recommended), open-frame,
AMS or CFS	Not Compatible
Compatible Nozzle Material	Hardened steel, diamond, tungsten carbide, etc
Compatible Nozzle Size	0.6 mm (recommended) / 0.4 mm / 0.8 mm
Compatible Plate Type	Smooth / Textured PEI Plate or other plate
Plate Surface Preparation	Apply liquid glue or PVP solid glue stick if the first layer bonding is not firm enough
Nozzle Temperature	280 - 300 °C
Bed Temperature	100 - 120 °C
Fan Speed	0% - 40%
Max Print Speed	250 mm/s

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